

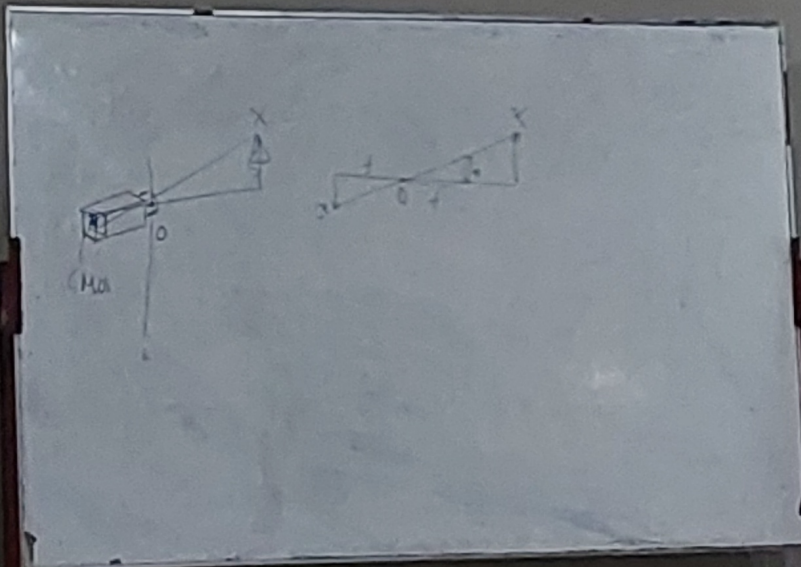
Epipolar Geometry

Slides are from Svetlana Lazebnik

Note in Prof. Lazebnik's slides: Many slides adapted from [J. Johnson and D. Fouhey](#)

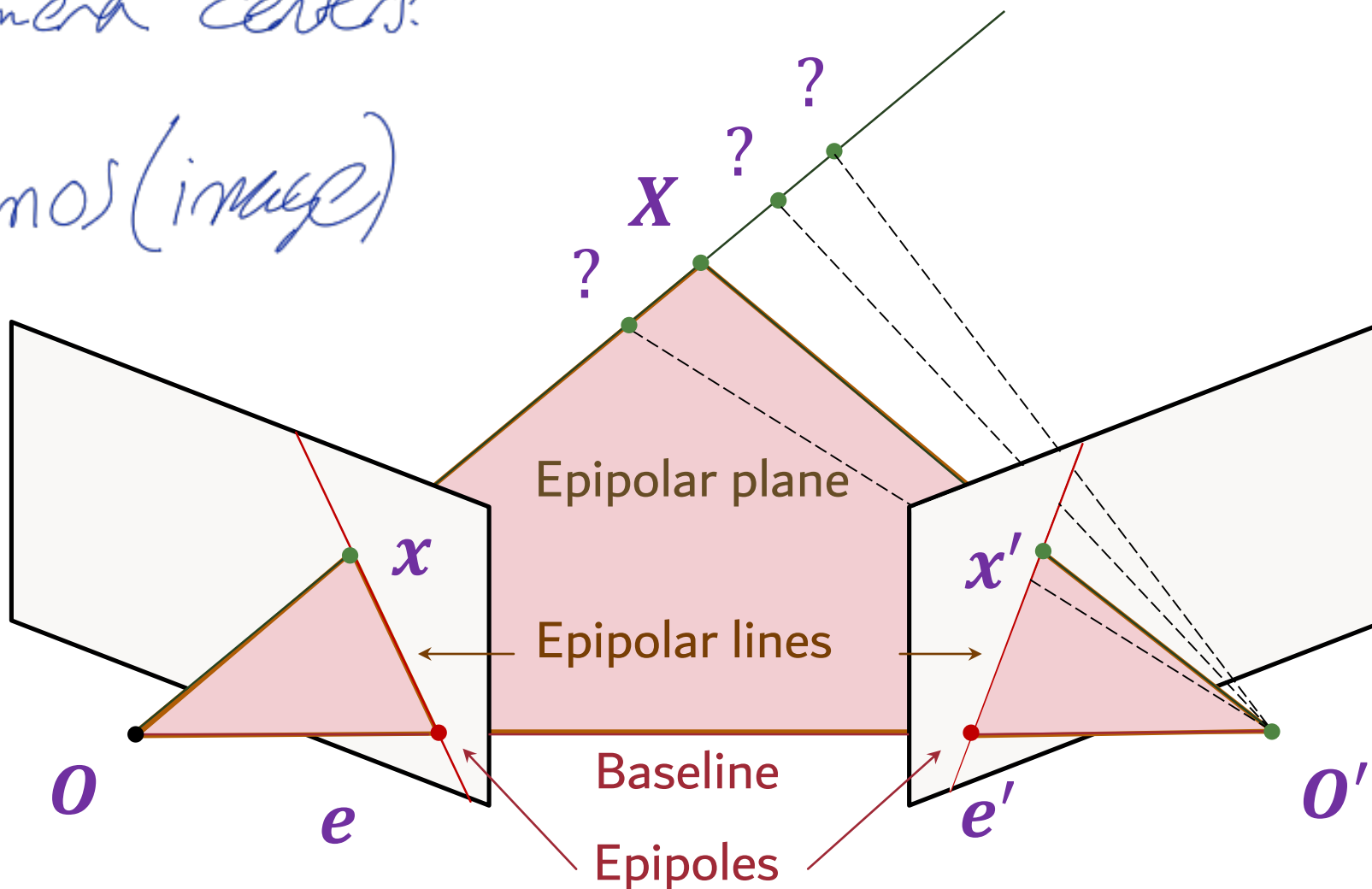
why image in front of camera

$$\frac{\alpha_2}{\alpha_1} = \frac{1}{2}$$
$$\alpha_2 = \frac{1}{2} \alpha_1$$
$$l = x_1 \times x_2$$
$$\begin{cases} 1x - 2 = 0 \\ 1y - 3 = 0 \end{cases}$$
$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \end{bmatrix}$$
$$\begin{array}{l} (0, 2) \\ (3, 0) \end{array}$$



O, O' camera centers:

$cmos(image)$



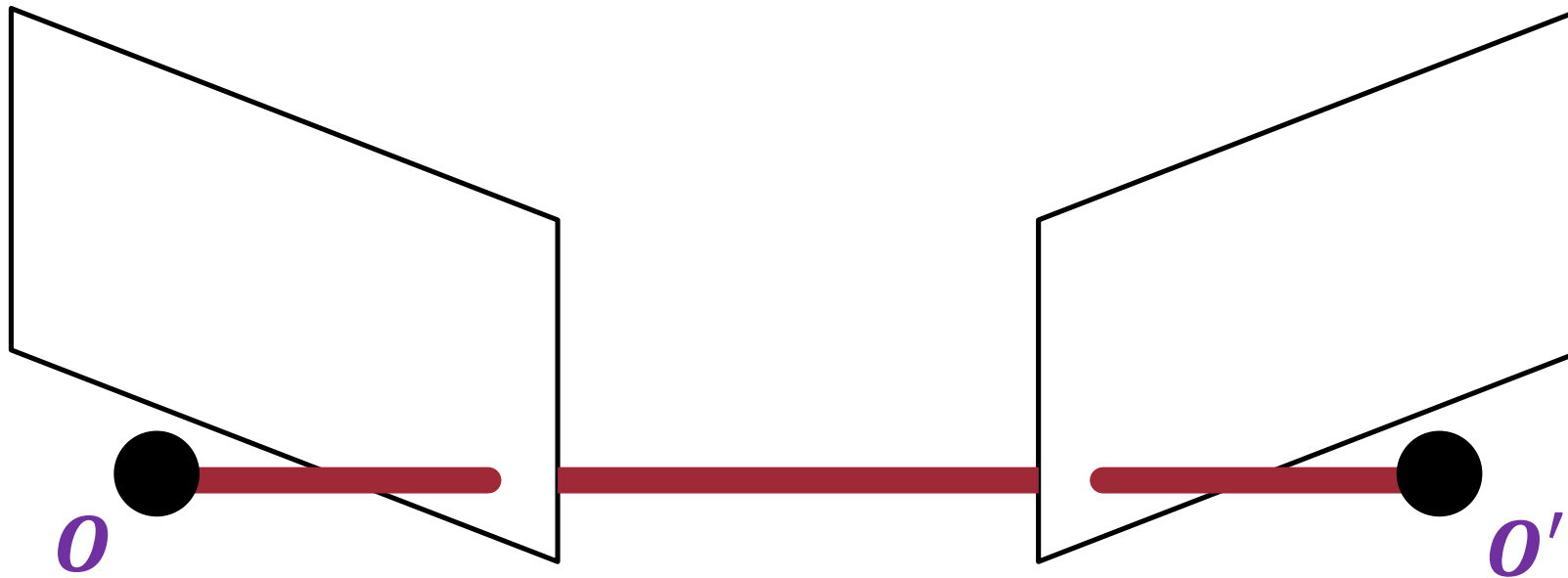
Left view

Right view

Outline

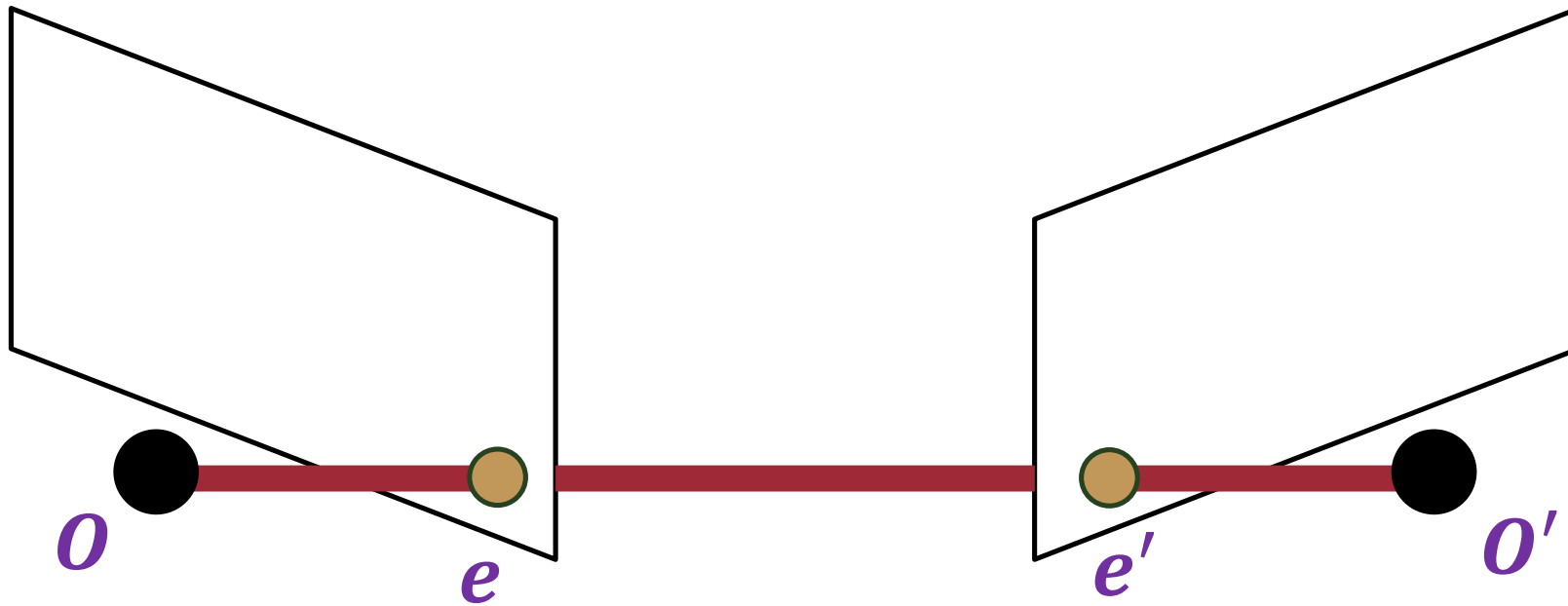
- Epipolar geometry setup
- Epipolar constraint
- Essential matrix
- Fundamental matrix
- Estimating the fundamental matrix

Epipolar Geometry Setup



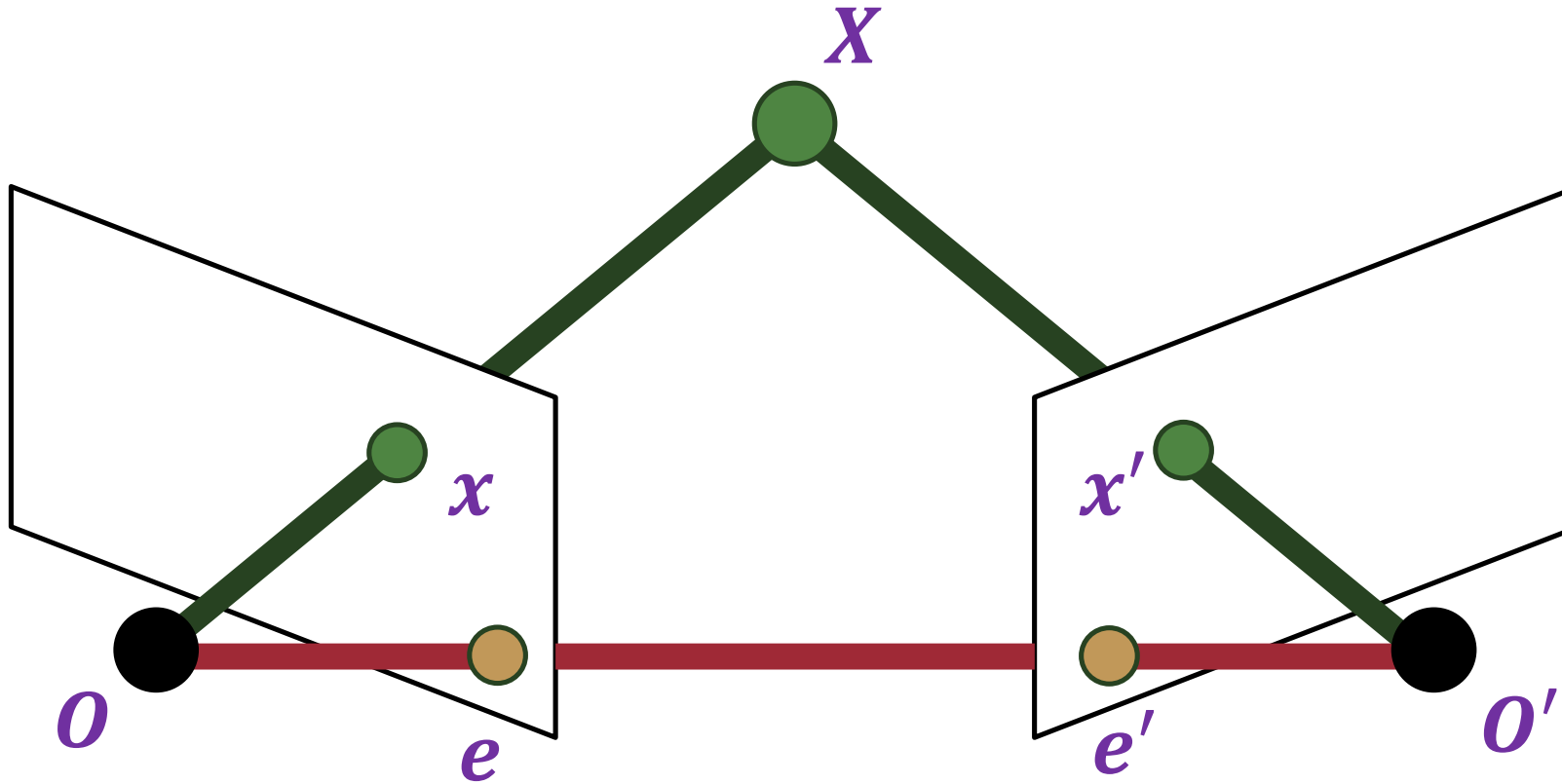
- Suppose we have two cameras with centers O, O'
- The **baseline** is the line connecting the origins

Epipolar Geometry Setup



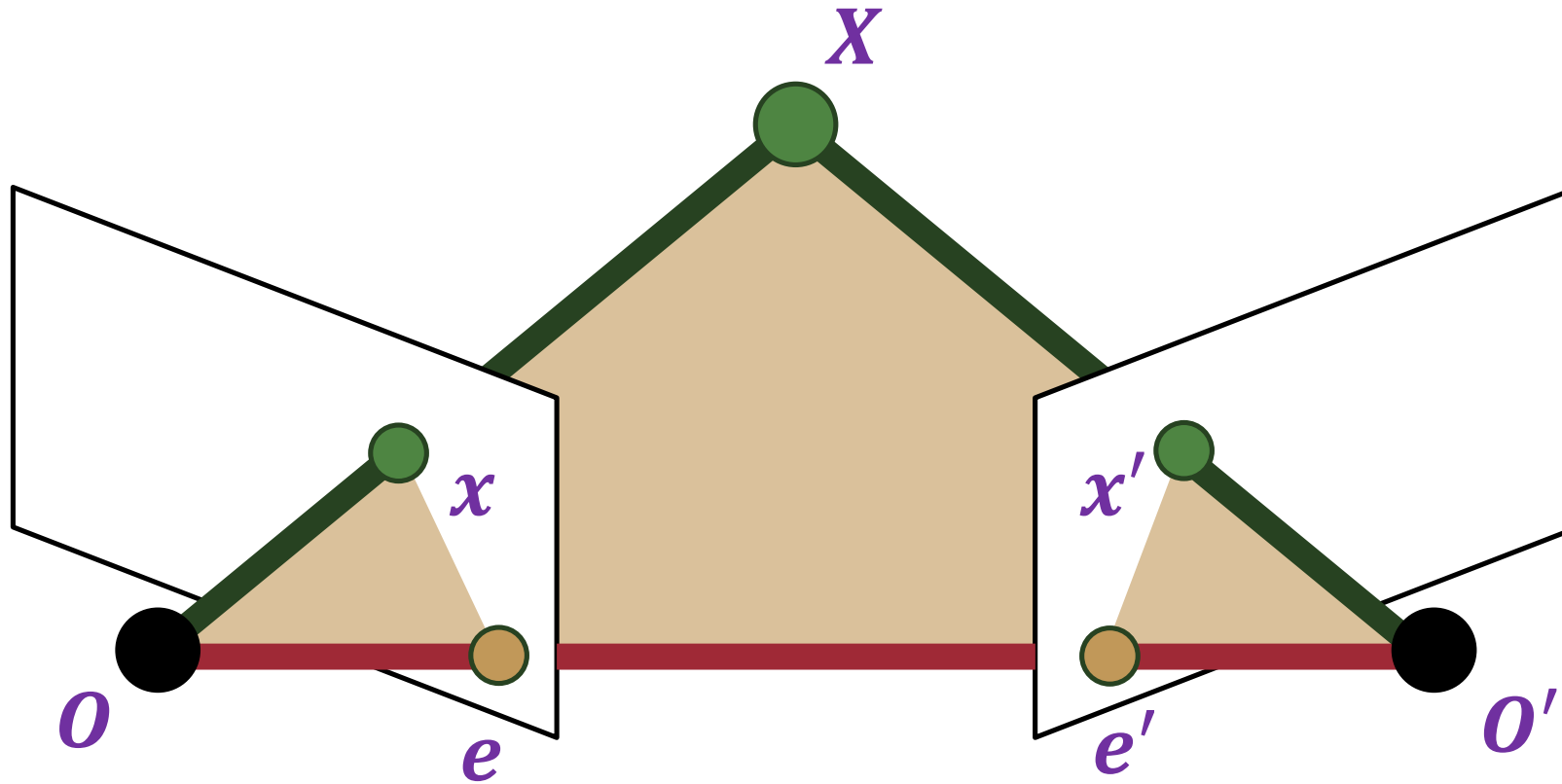
- **Epipoles** e, e' are where the baseline intersects the image planes, or projections of the other camera in each view

Epipolar Geometry Setup



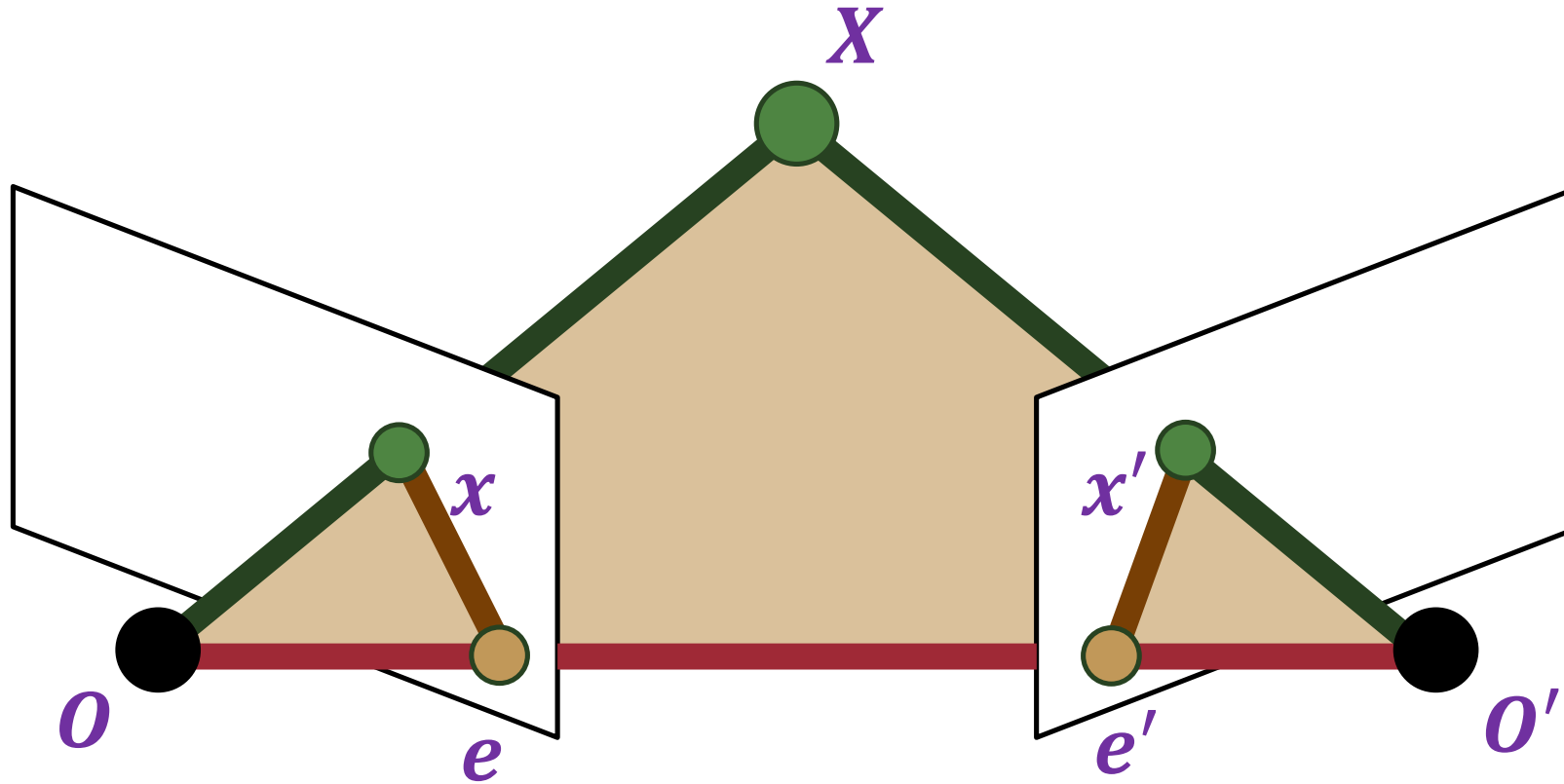
- Consider a **point** X , which projects to x and x'

Epipolar Geometry Setup



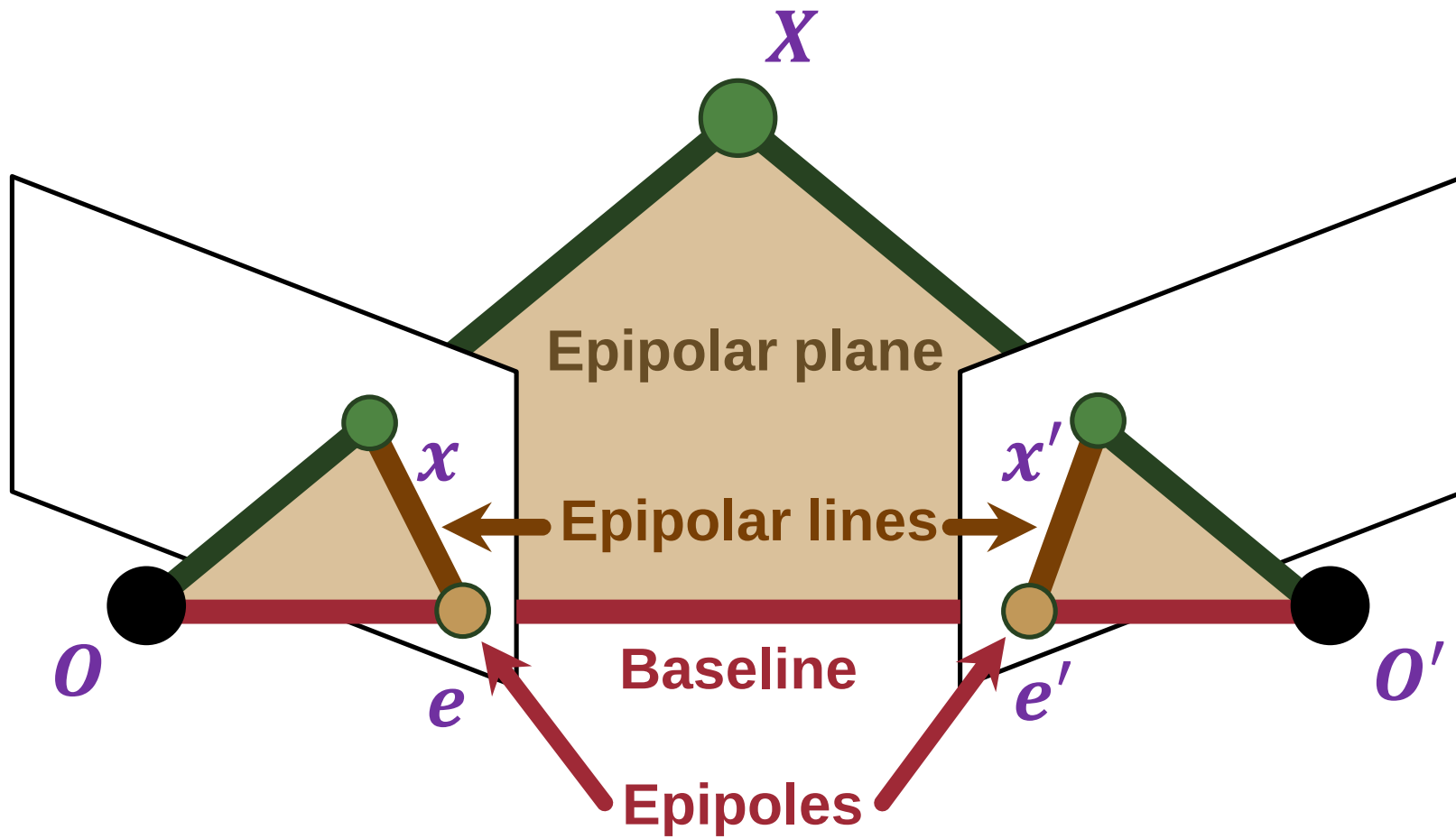
- The plane formed by X , O , and O' is called an **epipolar plane**
- There is a family of planes passing through O and O'

Epipolar Geometry Setup



- **Epipolar lines** connect the epipoles to the projections of X
- Equivalently, they are intersections of the epipolar plane with the image planes – thus, they come in matching pairs

Epipolar Geometry Setup: Summary

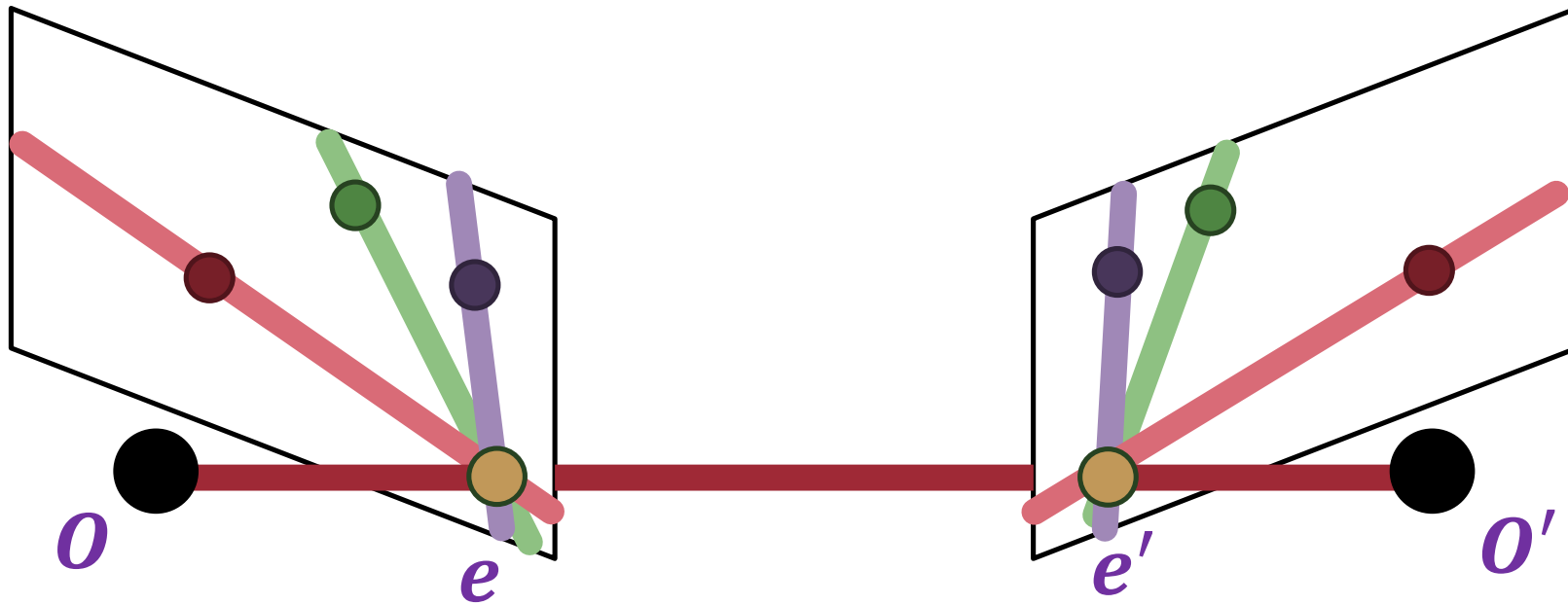


Example Configuration: Converging cameras

epipolar line go through the same point. (epipolar)

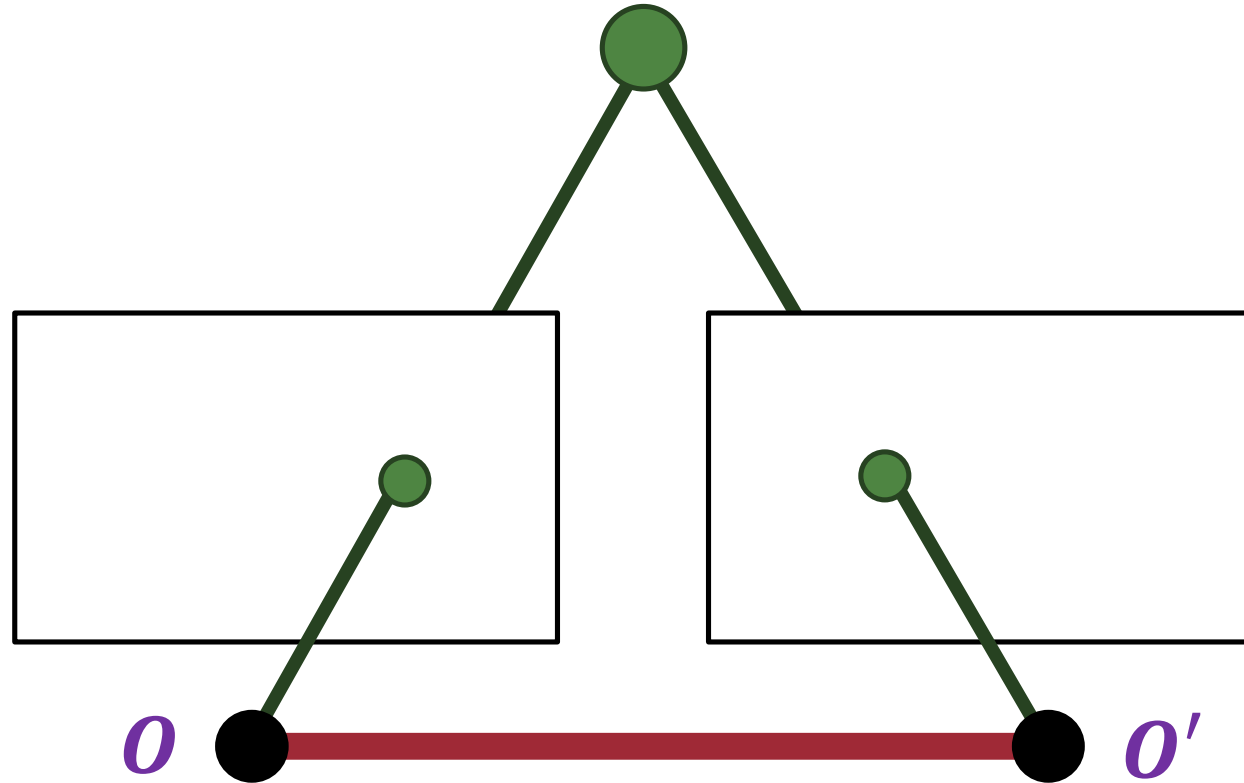


Example Configuration: Converging Cameras



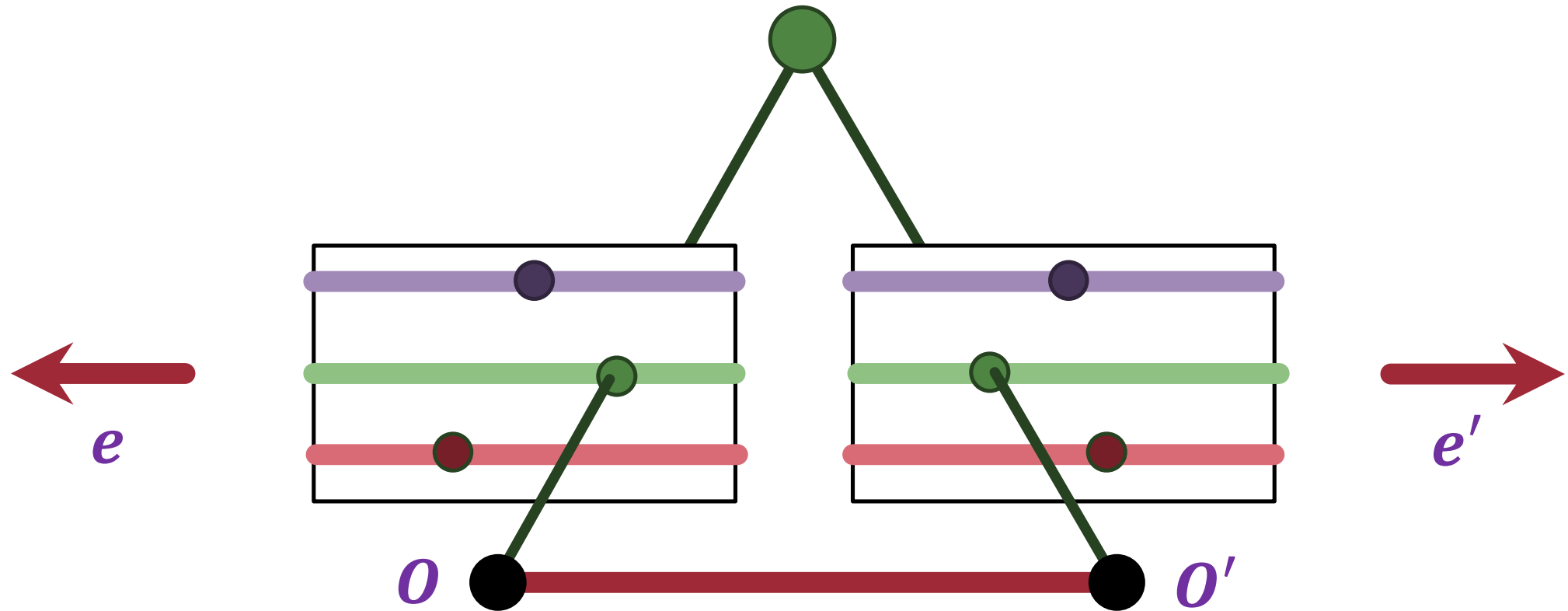
- Epipoles are finite, may be visible in the image

Example Configuration: Motion Parallel to Image Plane



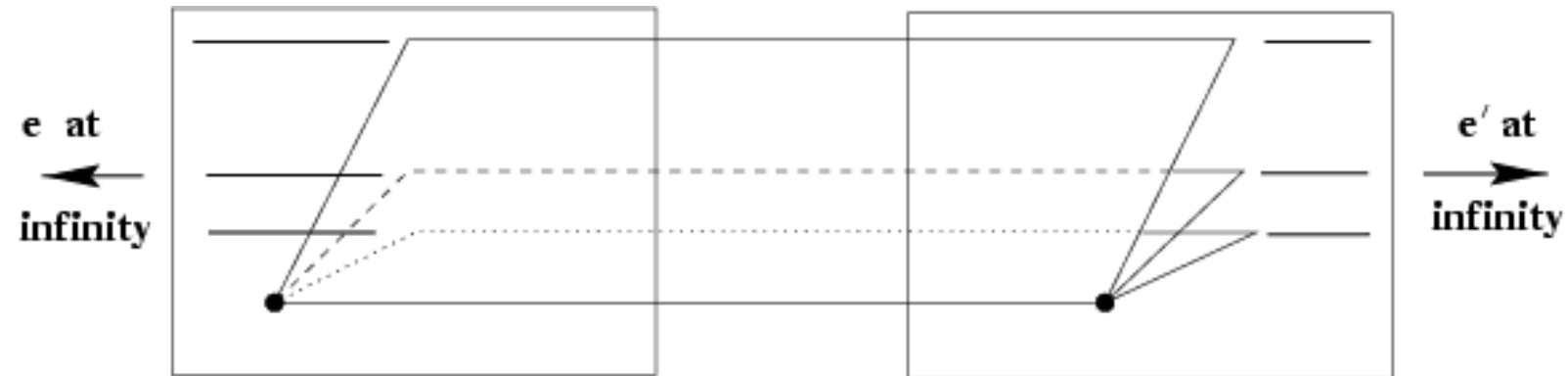
- Where are the epipoles and what do the epipolar lines look like?

Example configuration: Motion Parallel to Image Plane



- Epipoles *infinitely* far away, epipolar lines parallel

Example Configuration: Motion Parallel to Image Plane



Example configuration: Motion Perpendicular
to Image Plane

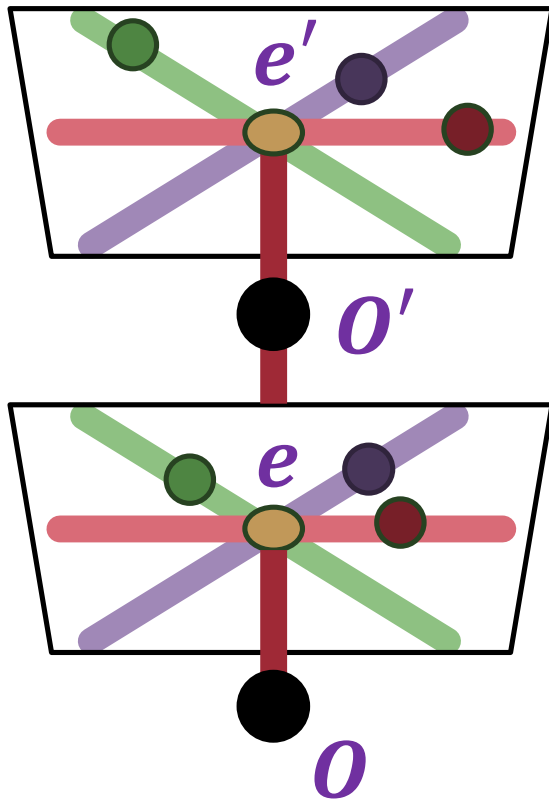


Example configuration: Motion Perpendicular
to Image Plane



Example Configuration: Motion Perpendicular to Image Plane

directing towards center

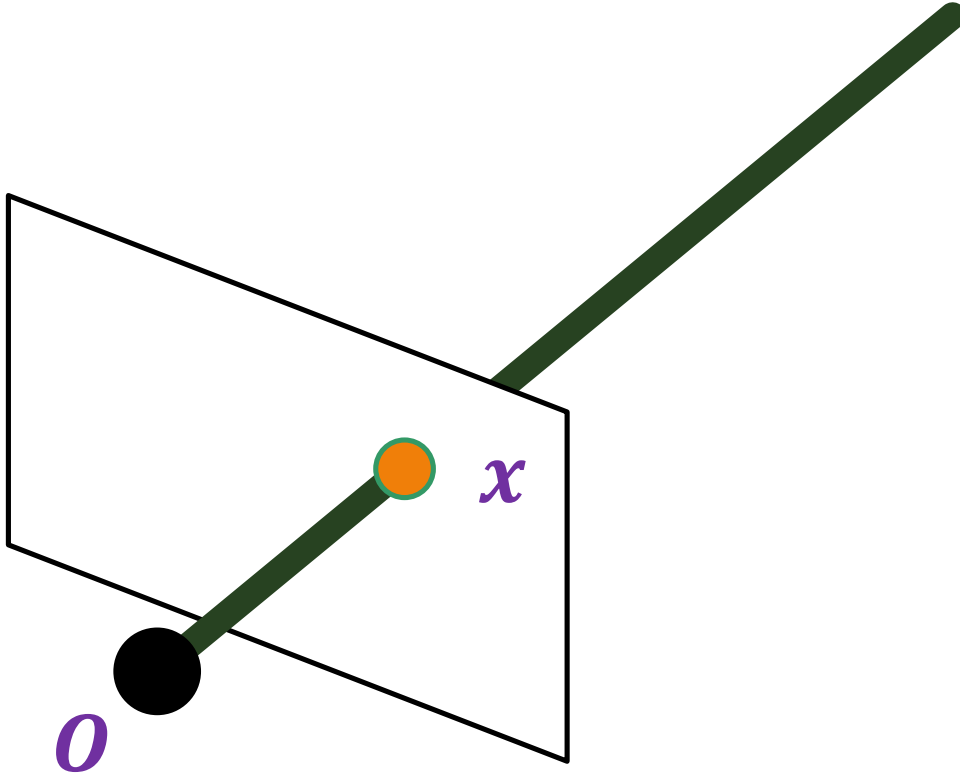


- Epipole is “focus of expansion” and coincides with the principal point of the camera
- Epipolar lines go out from principal point

Outline

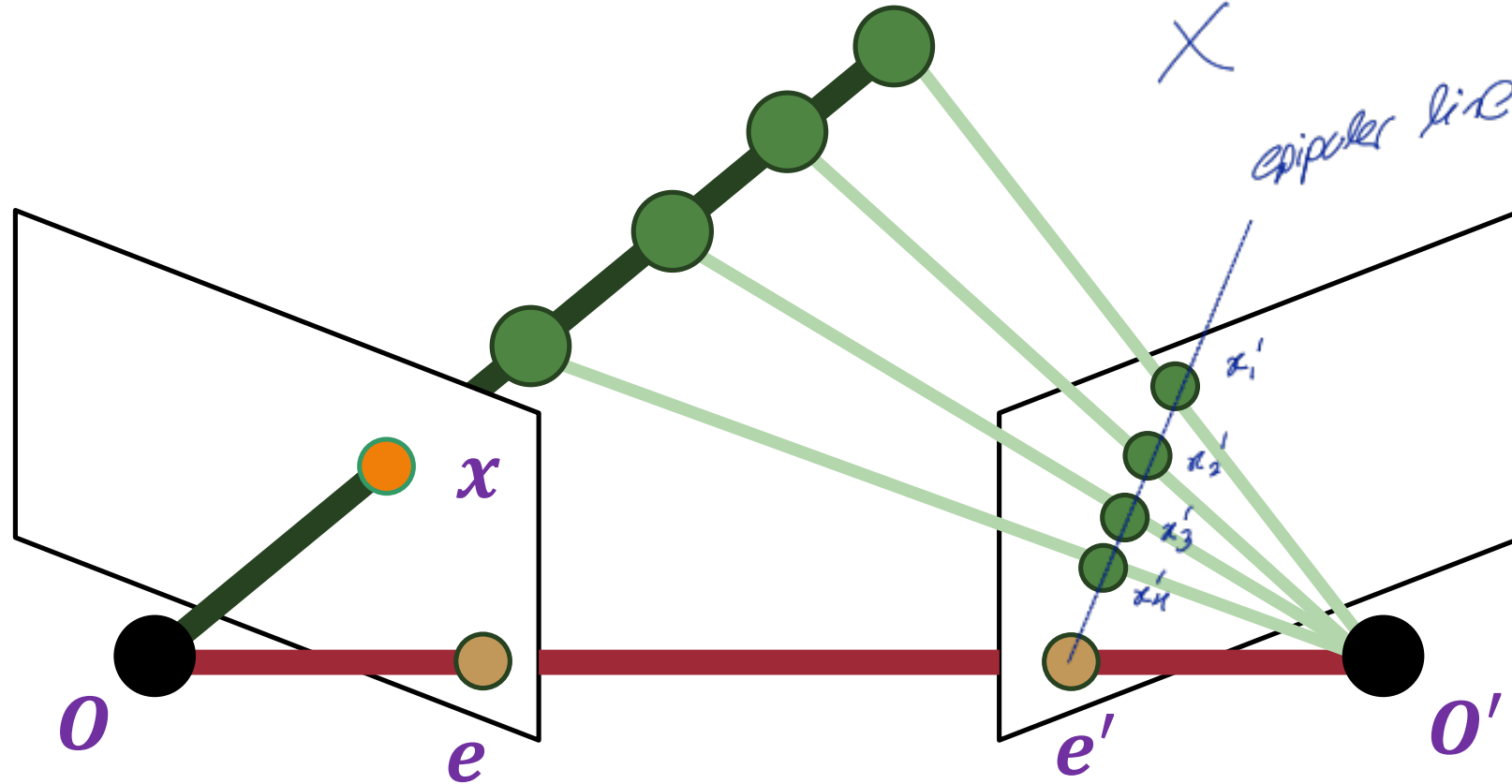
- Epipolar geometry setup
- Epipolar constraint

Epipolar Constraint



- Suppose we observe a single point x in one image

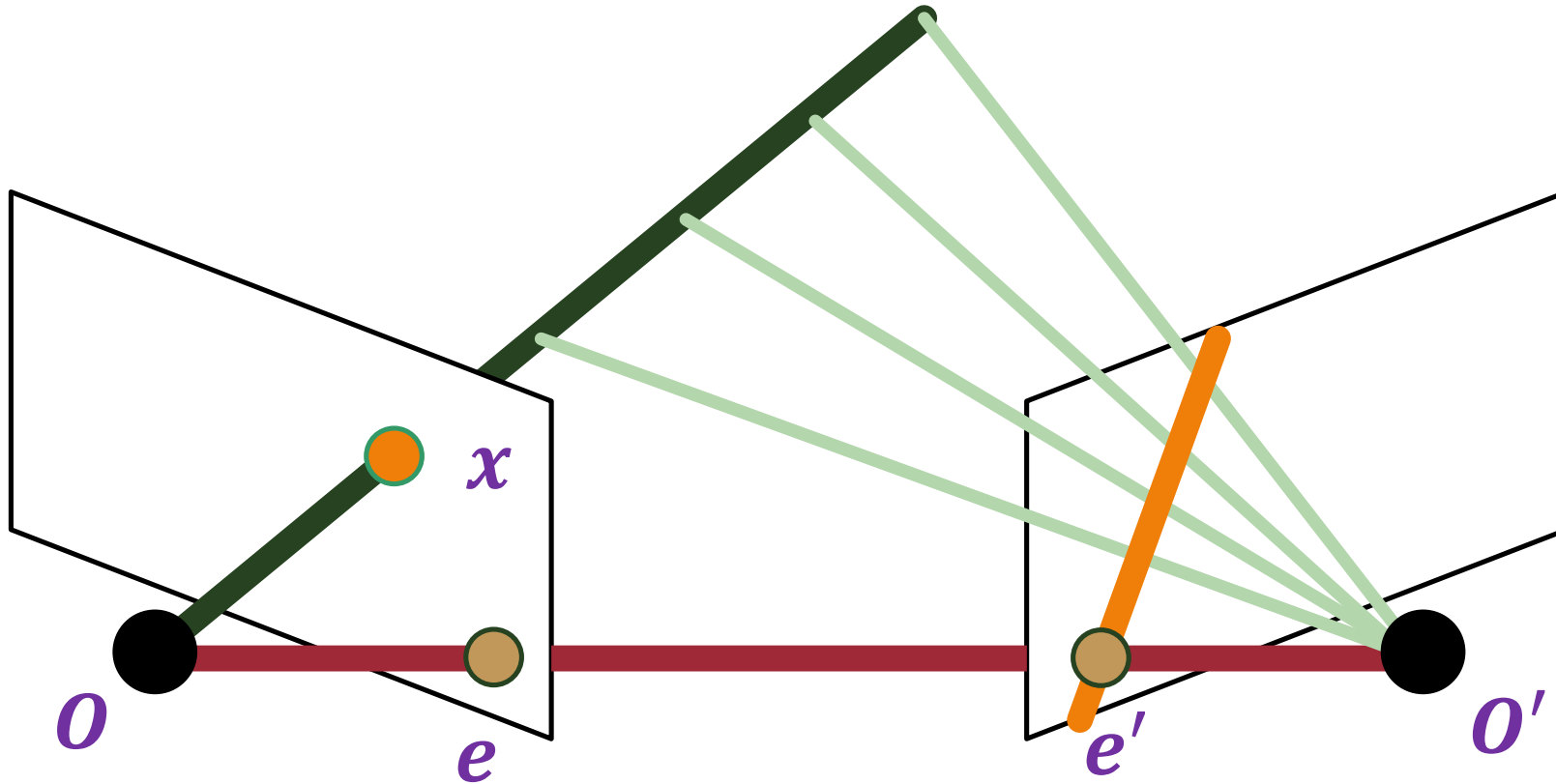
Epipolar constraint



- Where can we find the x' corresponding to x in the other image?

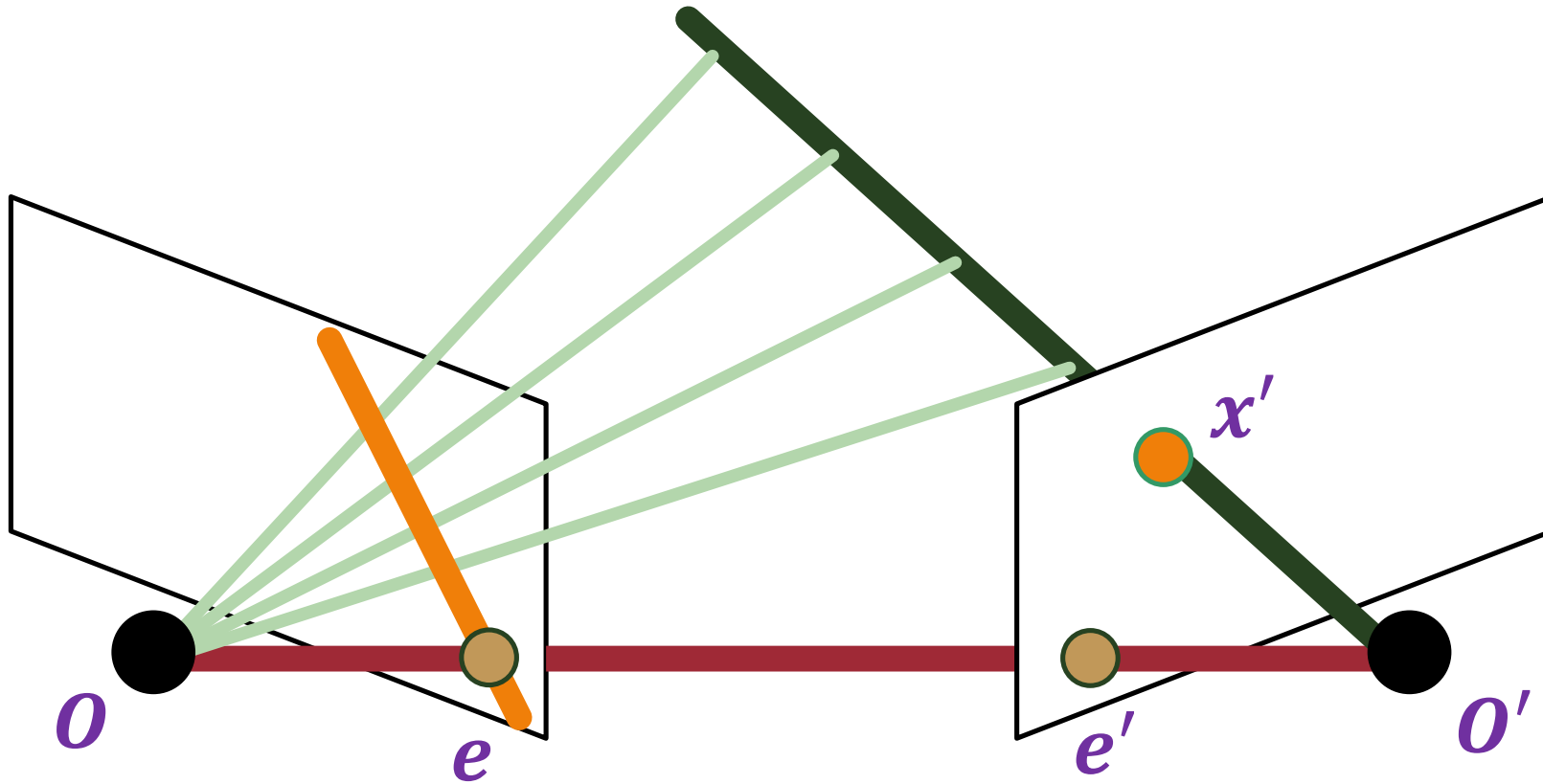
Epipolar Constraint

possible locations.



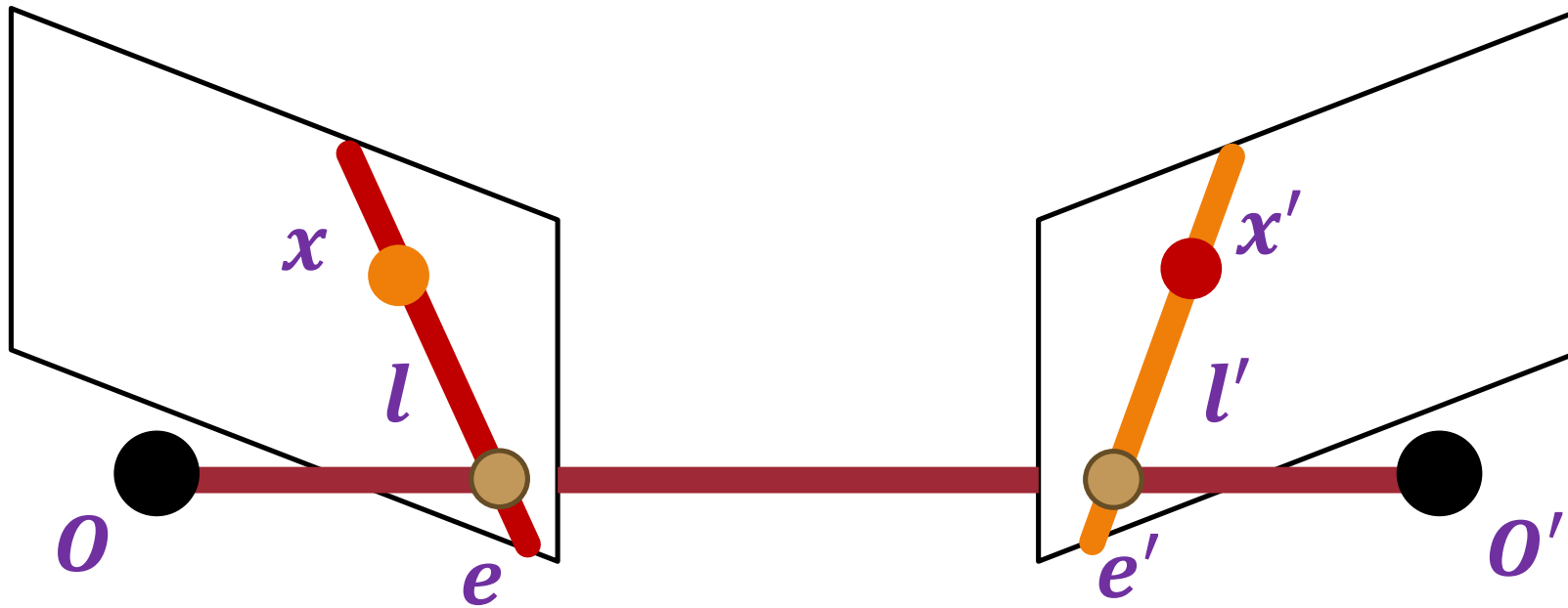
- Where can we find the x' corresponding to x in the other image?
- Along the **epipolar line** corresponding to x (projection of visual ray connecting O with x into the second image plane)

Epipolar Constraint



- Similarly, all points in the left image corresponding to x' have to lie along the epipolar line corresponding to x'

Epipolar constraint

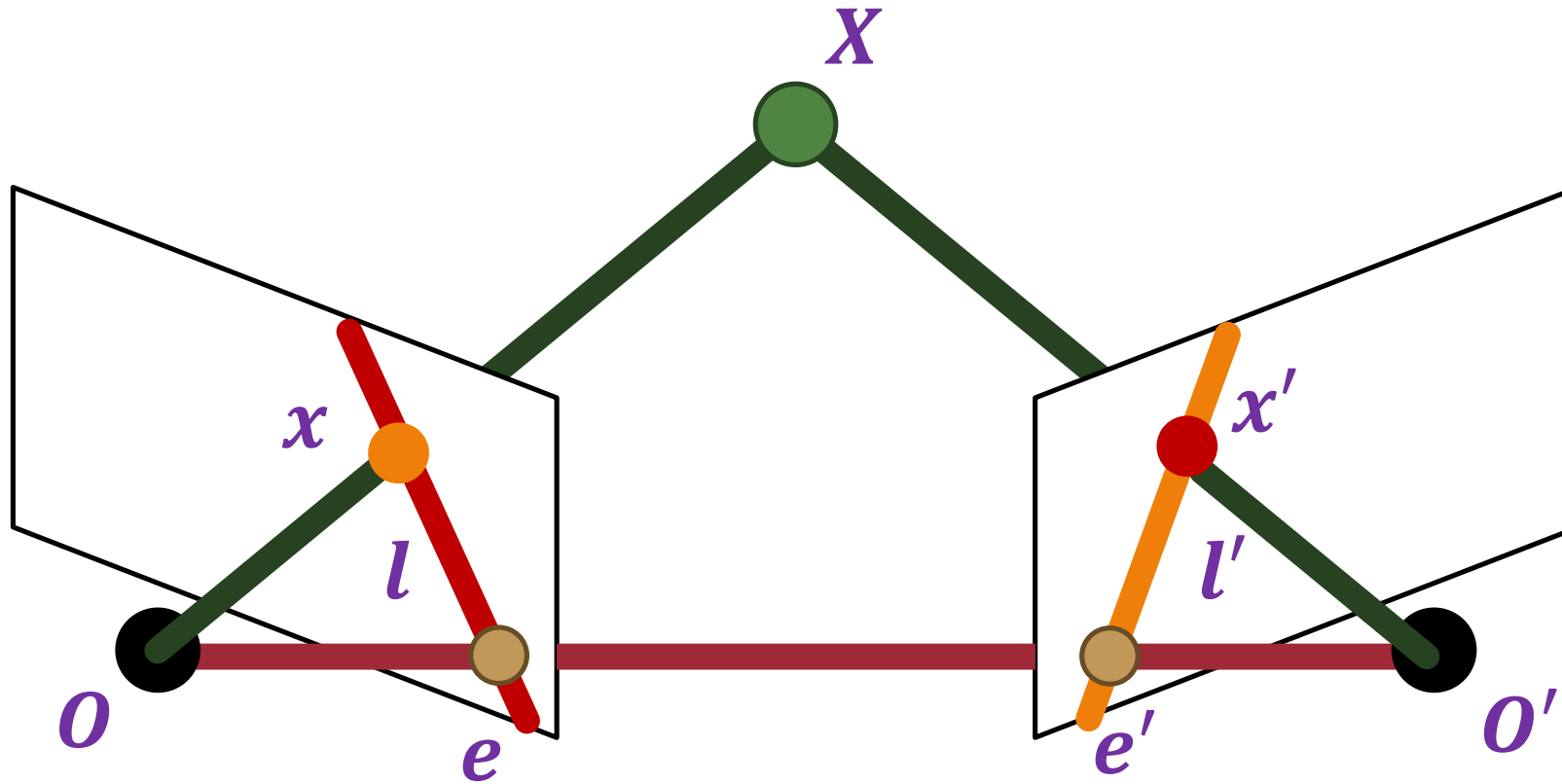


- Potential matches for x have to lie on the matching epipolar line l'
- Potential matches for x' have to lie on the matching epipolar line l

Epipolar Constraint: Example

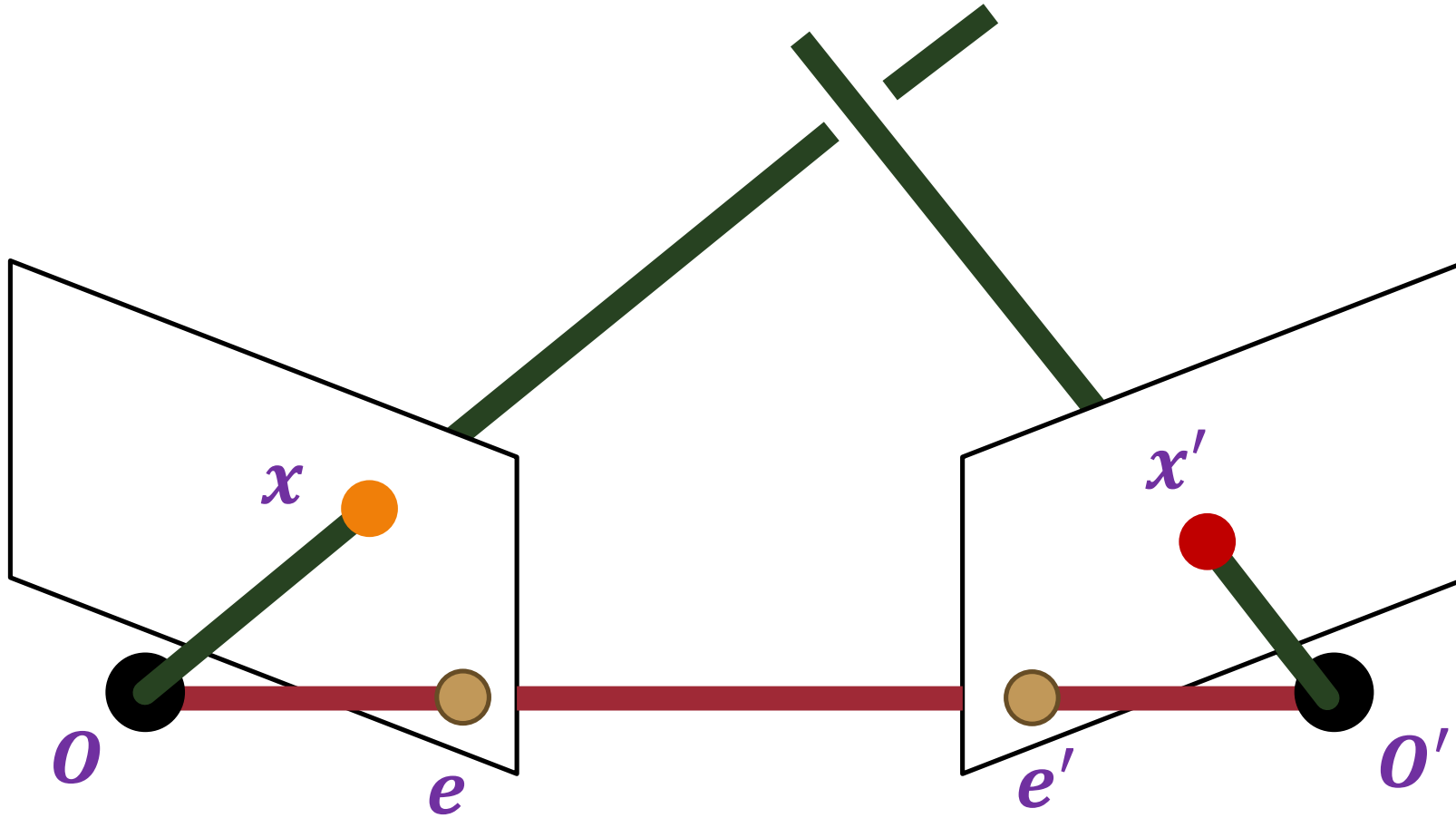


Epipolar constraint



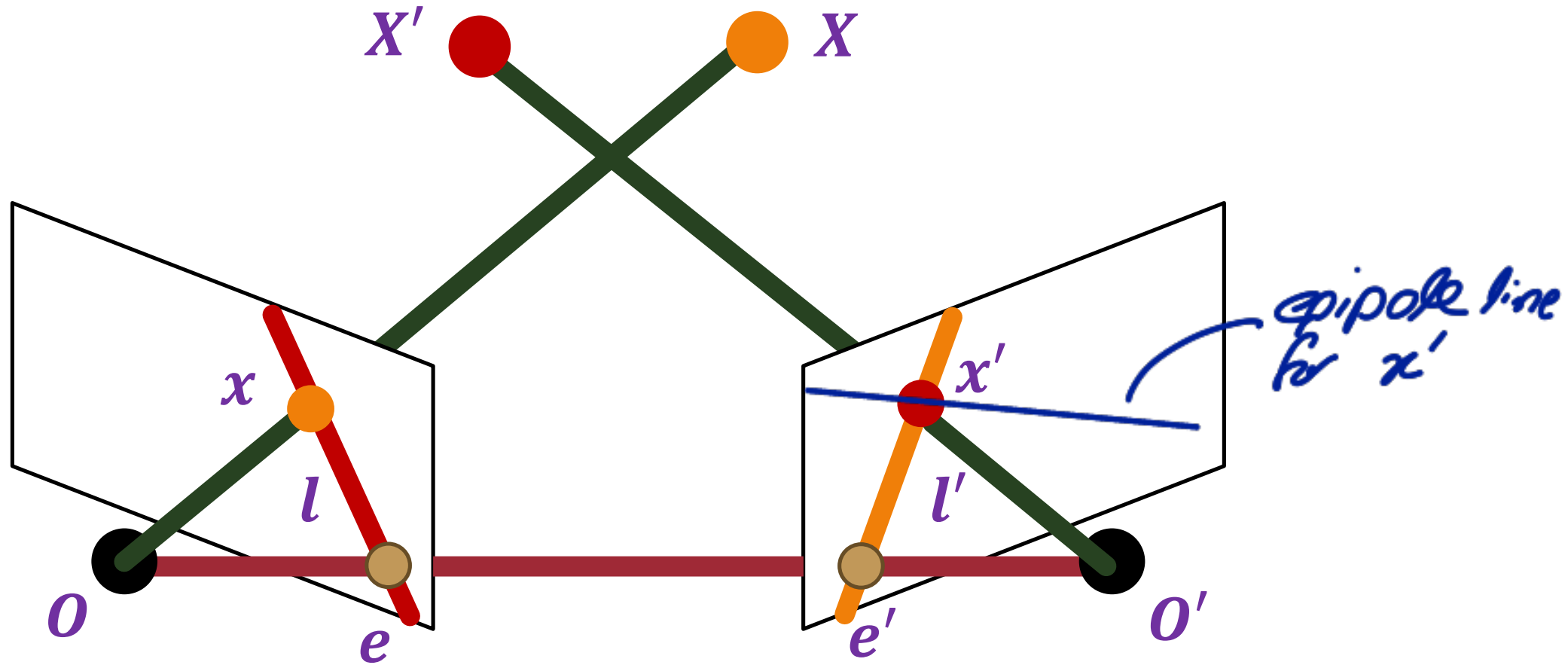
- Whenever two points x and x' lie on matching epipolar lines l and l' , the visual rays corresponding to them meet in space, i.e., x and x' could be projections of the same 3D point X

Epipolar Constraint



- Remember: in general, two rays *do not* meet in space!

Epipolar Constraint

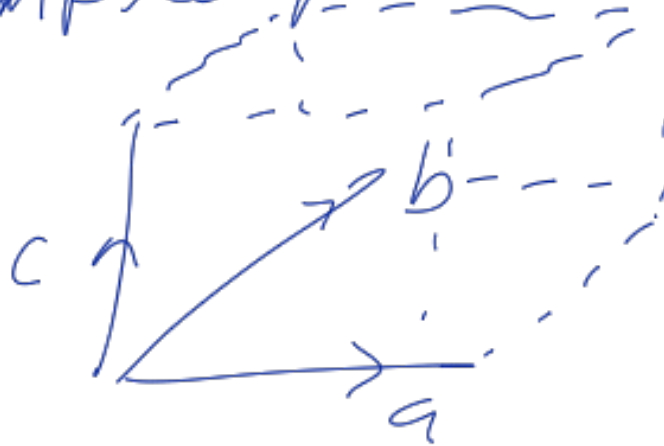


- Caveat: if x and x' satisfy the epipolar constraint, this doesn't mean they *have to be* projections of the same 3D point

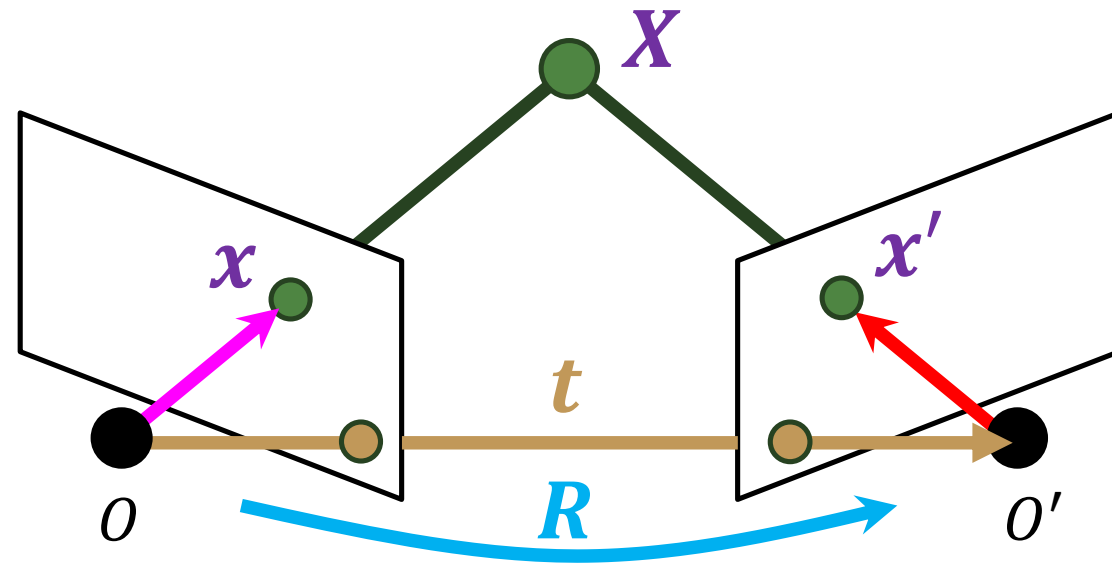
Outline

- Epipolar geometry setup
- Epipolar constraint
- Essential matrix

if 3 vectors are on the same plane, scalar triple product will be 0


$$\underline{c} \cdot (\underline{a} \times \underline{b}) = \text{Volume}$$

Math of the Epipolar Constraint: Calibrated Case



R is the rotation of camera O' with respect to camera O .
 t is the translation of camera O' with respect to camera O .

- Assume the intrinsic and extrinsic parameters of the cameras are known, world coordinate system is set to that of the first camera
- Then the projection matrices are given by $K[I \mid \mathbf{0}]$ and $K'[R \mid t]$ no rot no trans rot and trans to get O'
- We can pre-multiply the projection matrices (and the image points) by the inverse calibration matrices to get *normalized* image coordinates:

$$\mathbf{x}_{\text{norm}} = K^{-1}\mathbf{x}_{\text{pixel}} \equiv [I \mid \mathbf{0}]X, \quad \mathbf{x}'_{\text{norm}} = K'^{-1}\mathbf{x}'_{\text{pixel}} \equiv [R \mid t]X \leftarrow \text{in norm}$$

Recap

$$h x = K [R | t] X$$

↑ point in image wrt camera

3D point wrt camera

3D point wrt world

now since world is camera 1 origin

for cam 1 $h x = K_1 [I | 0] X$

for cam 2 $h x = K_2 [R | t] X$

now we notation

$$h x_{\text{pixel}} = K x_{\text{norm}}$$

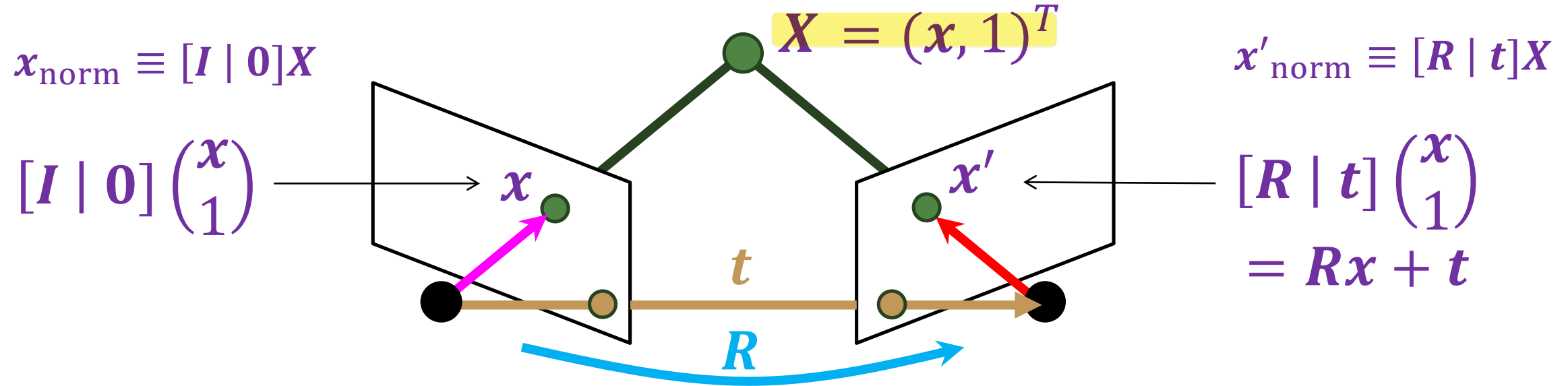
↑ point in image wrt camera

↑ 3D point wrt camera

$$x_{\text{norm}} = [R | t] X$$

Math of the Epipolar constraint: Calibrated case

when world and cam same



→ same plane.

- We have $x' \equiv Rx + t$
- This means the three vectors x' , Rx , and t are linearly dependent
- This constraint can be written using the *triple product*

$$x' \cdot [t \times (Rx)] = 0$$

So we have

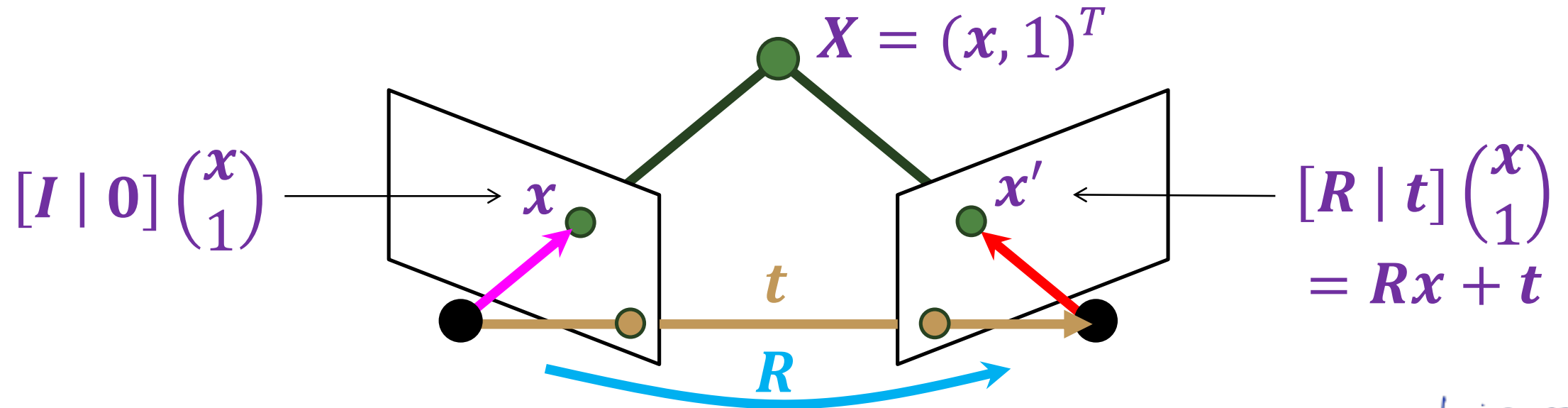
$$x_{\text{norm}} = [Z|0]X = x$$

$$\begin{aligned} x'_{\text{norm}} &= [R|t]x = Rx + t \\ &= Rx + t \end{aligned}$$

$$x'_{\text{norm}} = Rx + t \quad \left. \begin{array}{l} \text{Linear dependent} \\ \text{since on same plane} \\ \text{where } i \neq 0 \end{array} \right\}$$

$$x' \in [t \times R \times]$$

Math of the epipolar constraint: Calibrated case



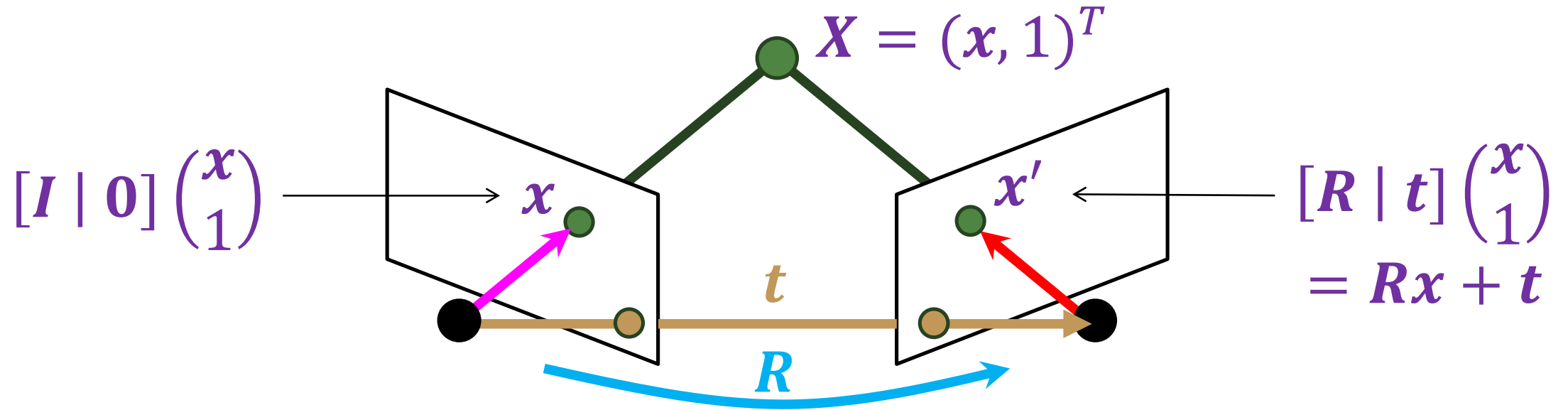
$$x' \cdot [t \times (Rx)] = 0$$

$$\Rightarrow x'^T [t_{\times}] Rx = 0$$

Write cross product in matrix
(skew symmetric matrix)

Recall: $a \times b = \underbrace{\begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}}_{[a_{\times}]} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = [a_{\times}] b$

Math of the epipolar constraint: Calibrated case

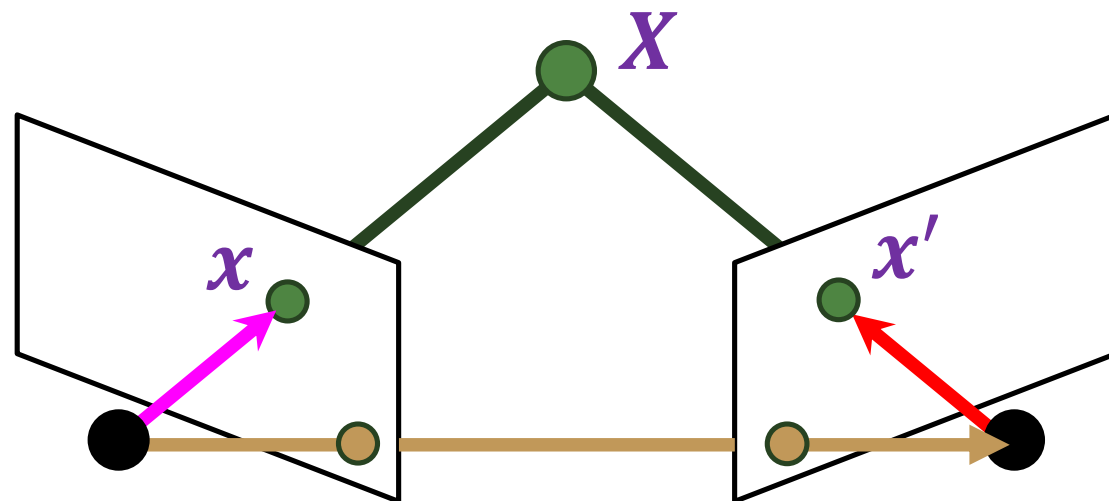


$$x' \cdot [t \times (Rx)] = 0 \quad \Rightarrow \quad x'^T [t_{\times}] Rx = 0 \quad \Rightarrow \quad x'^T E x = 0$$

$$E = [t_{\times}] R$$

Essential Matrix

The Essential Matrix



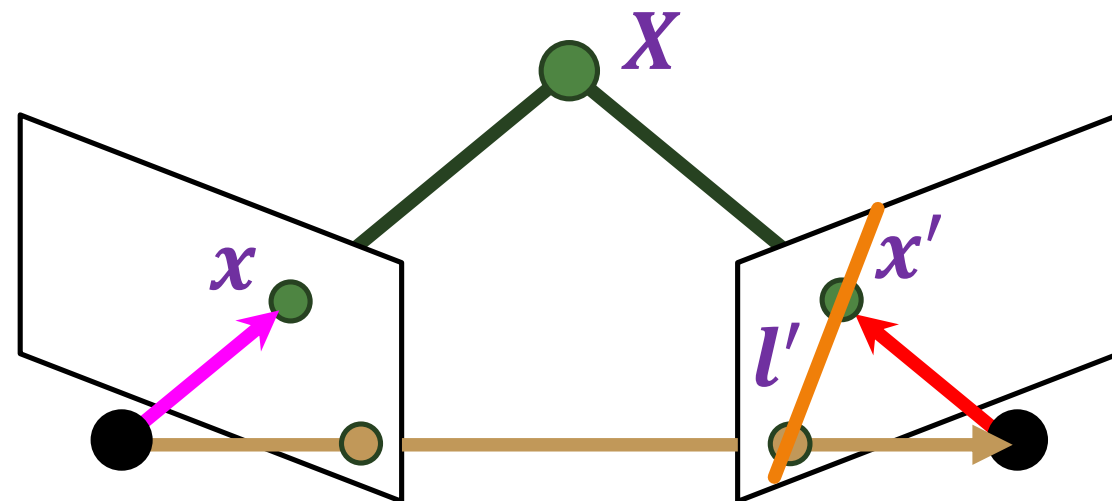
$$\mathbf{x}'^T \mathbf{E} \mathbf{x} = 0$$

normalized

$$(x', y', 1) \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = 0$$

x on line l
 $x'^T l = 0$
 so since x' is on
 $l' \quad Ex = l'$

The Essential Matrix: Properties



Always Capital $X_{(4 \times 1)}$
simple $\pi_{(3 \times 1)}$
last param = 1

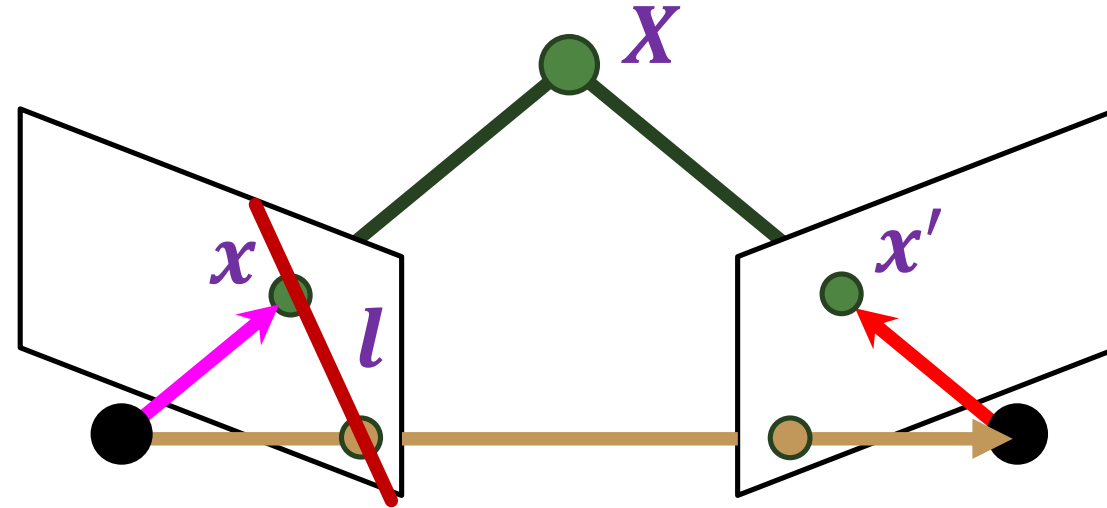
$$x'^T E x = 0$$

l' depend
on x'

- $E x$ is the epipolar line associated with x ($l' = E x$)

Recall: a line is given by $a x + b y + c = 0$ or $l^T x = 0$
where $l = (a, b, c)^T$ and $x = (x, y, 1)^T$

The Essential Matrix: Properties



epipolar line is l'
similar to $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$

by using x' point
 $x'^T l' = 0$

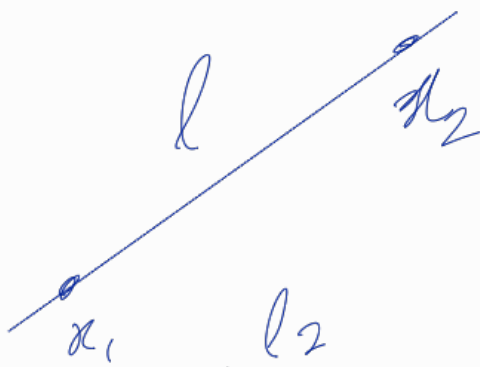
$$x'^T E x = 0$$

- $E x$ is the epipolar line associated with x ($l' = E x$)
- $E^T x'$ is the epipolar line associated with x' ($l = E^T x'$)
- $E e = 0$ and $E^T e' = 0$
- E is singular (rank two) and has five degrees of freedom

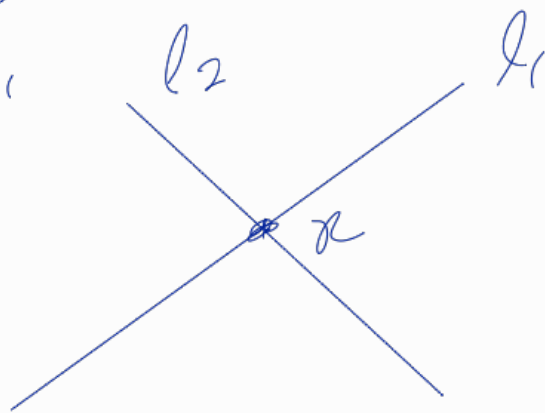
Do this

Remember

important property



$$l = x_1 \times x_2$$



$$x = l_1 \times l_2$$

However what we used as 3D
cam coordinates

But what we actually want is 2D
pixel coordinates

$$x_1'^T E x_2 = 0$$

$$x_p = K x_n$$

$$x'_p = K' x'_n$$

$$(K'^{-1} x'_p)^T E (K^{-1} x_p) = 0$$

$$x'_p{}^T (K')^{-T} E K^{-1} x_p = 0$$

$$F = (K')^{-T} E K^{-1}$$

← epipolar line
wrt the image
coord
what we actually
need

Outline

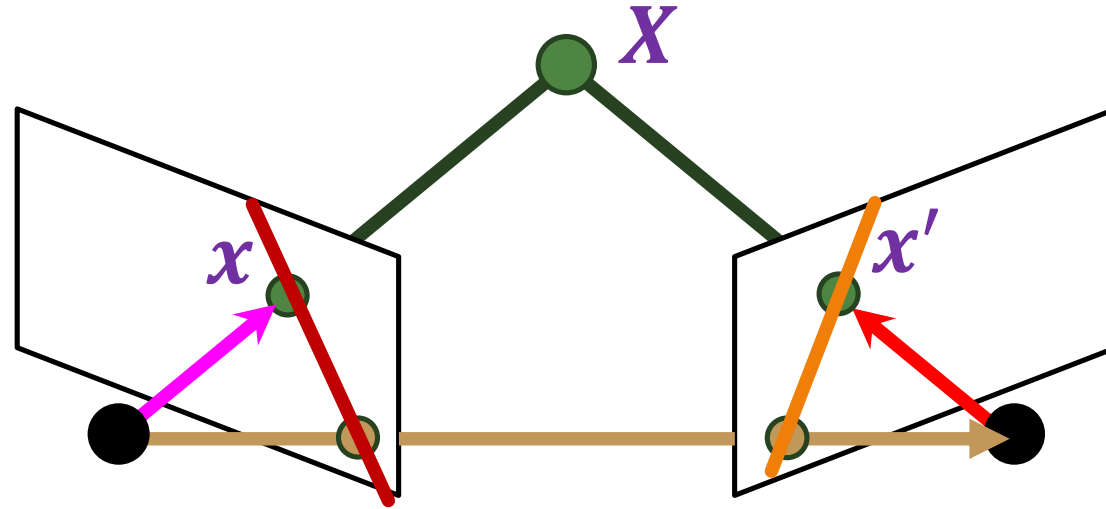
- Epipolar geometry setup
- Epipolar constraint
- Essential matrix
- Fundamental matrix

Back
multiple view
geometry

after coords normalized,

when we don't know
cam cali matrix?

Epipolar Constraint: Uncalibrated Case

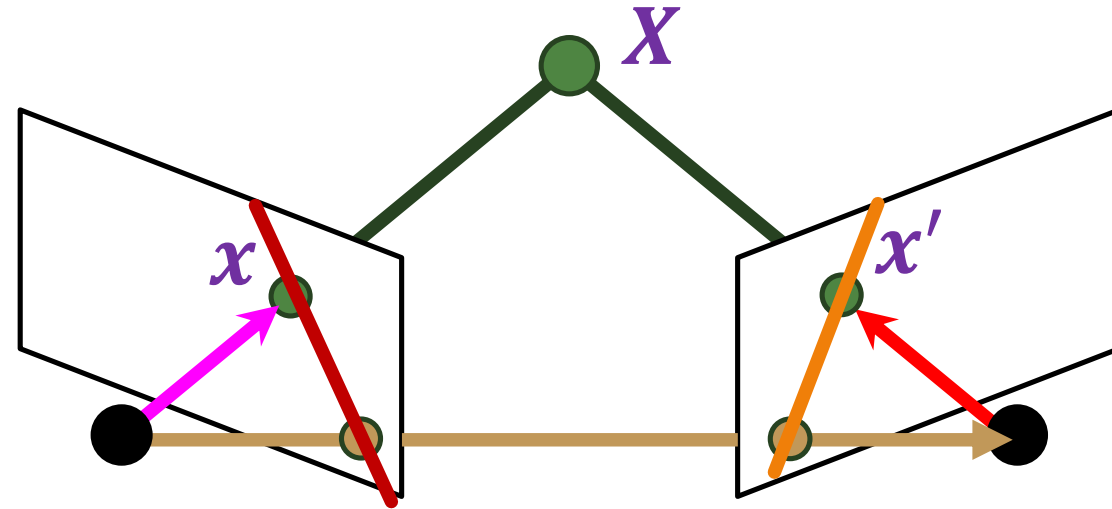


- The calibration matrices K and K' of the two cameras are unknown
- We can write the epipolar constraint in terms of *unknown* normalized coordinates:

$$\mathbf{x}_{\text{norm}}'^T \mathbf{E} \mathbf{x}_{\text{norm}} = 0,$$

where $\mathbf{x}_{\text{norm}} = K^{-1} \mathbf{x}$, $\mathbf{x}_{\text{norm}}' = K'^{-1} \mathbf{x}'$

Epipolar Constraint: Uncalibrated Case



*no need for
normalized
images*

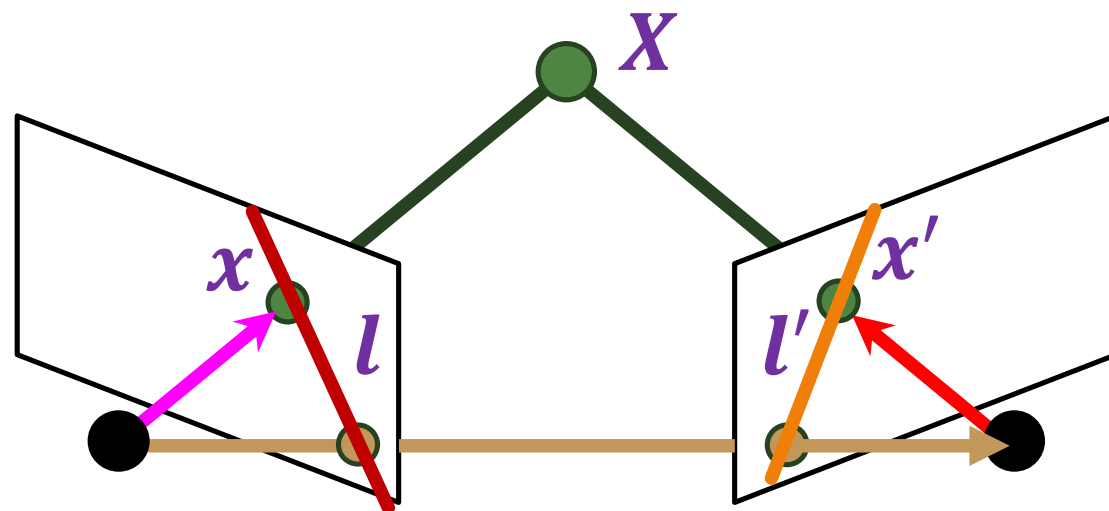
$$\mathbf{x}_{\text{norm}}'^T \mathbf{E} \mathbf{x}_{\text{norm}} = 0 \quad \Rightarrow \quad \mathbf{x}'^T \mathbf{F} \mathbf{x} = 0, \text{ where } \mathbf{F} = \mathbf{K}'^{-T} \mathbf{E} \mathbf{K}^{-1}$$

$$\mathbf{x}_{\text{norm}} = \mathbf{K}^{-1} \mathbf{x}$$

$$\mathbf{x}_{\text{norm}}' = \mathbf{K}'^{-1} \mathbf{x}'$$

Fundamental Matrix

The Fundamental Matrix

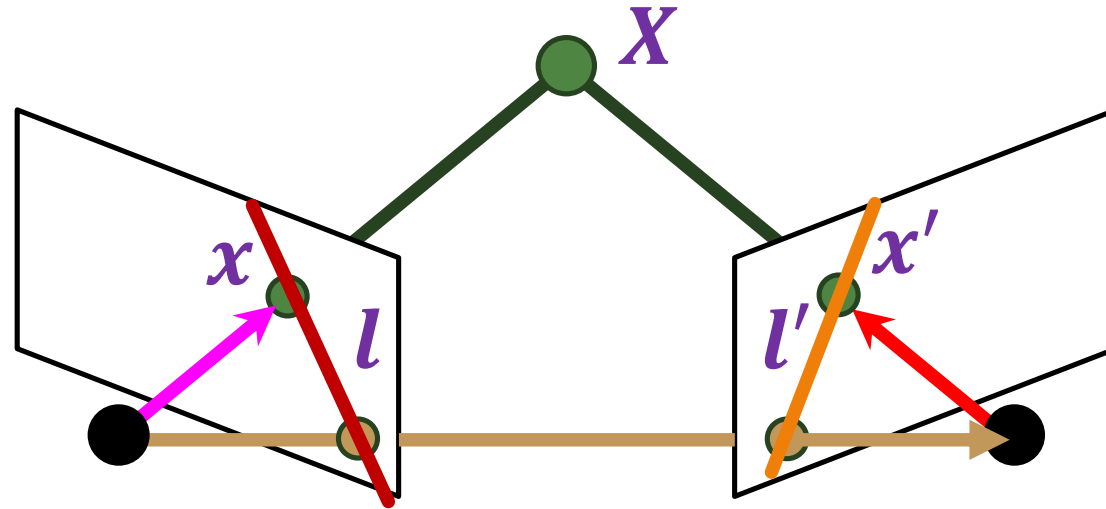


$$\mathbf{x}'^T \mathbf{F} \mathbf{x} = 0$$

image points in
2 cams are
related by $\mathbf{F}$

$$(x', y', 1) \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = 0$$

The Fundamental Matrix: Properties



$$\mathbf{x}'^T \mathbf{F} \mathbf{x} = 0$$

- $\mathbf{F} \mathbf{x}$ is the epipolar line associated with $\mathbf{x}$ ($\mathbf{l}' = \mathbf{F} \mathbf{x}$)
- $\mathbf{F}^T \mathbf{x}'$ is the epipolar line associated with $\mathbf{x}'$ ($\mathbf{l} = \mathbf{F}^T \mathbf{x}'$)
- $\mathbf{F} \mathbf{e} = 0$ and $\mathbf{F}^T \mathbf{e}' = 0$
- $\mathbf{F}$ is singular (rank two) and has **seven** degrees of freedom

Outline

- Epipolar geometry setup
- Epipolar constraint
- Essential matrix
- Fundamental matrix
- Estimating the fundamental matrix

Estimating the Fundamental Matrix

- Given: correspondences $\mathbf{x} = (x, y, 1)^T$ and $\mathbf{x}' = (x', y', 1)^T$
first need to find corresponding points



Estimating the Fundamental Matrix

- Given: correspondences $\mathbf{x} = (x, y, 1)^T$ and $\mathbf{x}' = (x', y', 1)^T$
- Constraint: $\mathbf{x}'^T \mathbf{F} \mathbf{x} = 0$

$$(x', y', 1) \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = 0 \quad \Rightarrow \quad (x'x, x'y, x'y', y'y, y', x, y, 1) \begin{pmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{pmatrix} = 0$$

This is for one point:

The Eight Point Algorithm

similar to $A^T A$

$$\underbrace{\begin{bmatrix} x'x & x'y & x' & y'x & y'y & y' & x & y & 1 \\ \vdots & & & & & & & & \\ \vdots & & & & & & & & \end{bmatrix}}_{\mathbf{U}} \begin{pmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{pmatrix} = \mathbf{0}$$

$f=0$
trivial $\times$

- Homogeneous least squares to find $\mathbf{f}$:

$$\arg \min_{\|\mathbf{f}\|=1} \|\mathbf{U}\mathbf{f}\|_2^2 \rightarrow \text{Eigenvector of } \mathbf{U}^T \mathbf{U} \text{ with smallest eigenvalue}$$

Enforcing Rank-2 Constraint

- We know F needs to be singular/rank 2. How do we force it to be singular?
- Solution: take SVD of the initial estimate and throw out the smallest singular value

$$F_{\text{init}} = U\Sigma V^T$$



$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$$



$$\Sigma' = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



$$F = U\Sigma'V^T$$

Enforcing Rank-2 Constraint

Initial F estimate



after Rank-2 estimate



Normalized Eight Point Algorithm

???

Divide all
coords by
largest no.
to
normalize-

$$\begin{array}{ccccccc}
 & 10^6 & 10^6 & 10^3 & 10^6 & 10^6 & 10^3 & 10^3 & 10^3 & 1 \\
 & \swarrow \text{since multiplied} & & & & & & & & \\
 \underbrace{\begin{bmatrix} \vdots & & & & & & & & & \\ x'x & x'y & x' & y'x & y'y & y' & x & y & 1 & \\ \vdots & & & & & & & & & \end{bmatrix}}_{\mathbf{U}} & \begin{pmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{pmatrix} & = \mathbf{0}
 \end{array}$$

- Recall that x, y, x', y' are pixel coordinates. What might be the order of magnitude of each column of $\mathbf{U}$?
- This causes numerical instability!

$U^T U$ find eigen
vector of smallest
eigen value.

The Normalized Eight-Point Algorithm

- In each image, center the set of points at the origin, and scale it so the mean squared distance between the origin and the points is 2 pixels
- Use the eight-point algorithm to compute F from the normalized points
- Enforce the rank-2 constraint
- Transform fundamental matrix back to original units: if T and T' are the normalizing transformations in the two images, then the fundamental matrix in original coordinates is $T'^T F T$

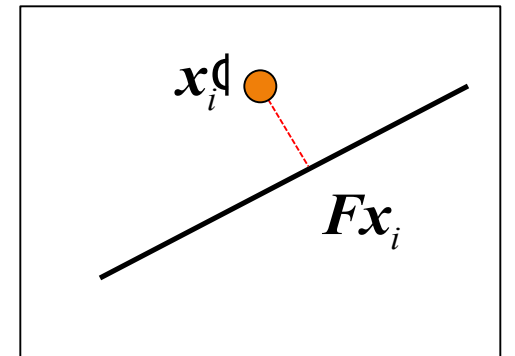
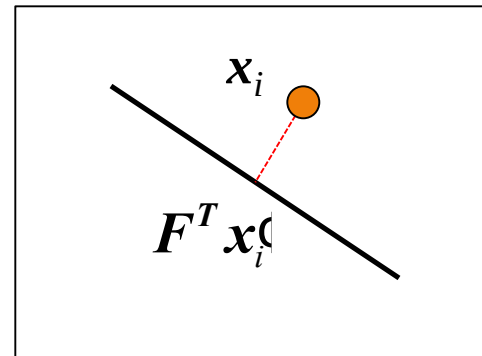
Nonlinear Estimation

- Linear estimation minimizes the sum of squared *algebraic* distances between points $\mathbf{x}'_i$ and epipolar lines $\mathbf{F}\mathbf{x}_i$ (or points $\mathbf{x}_i$ and epipolar lines $\mathbf{F}^T\mathbf{x}'_i$):

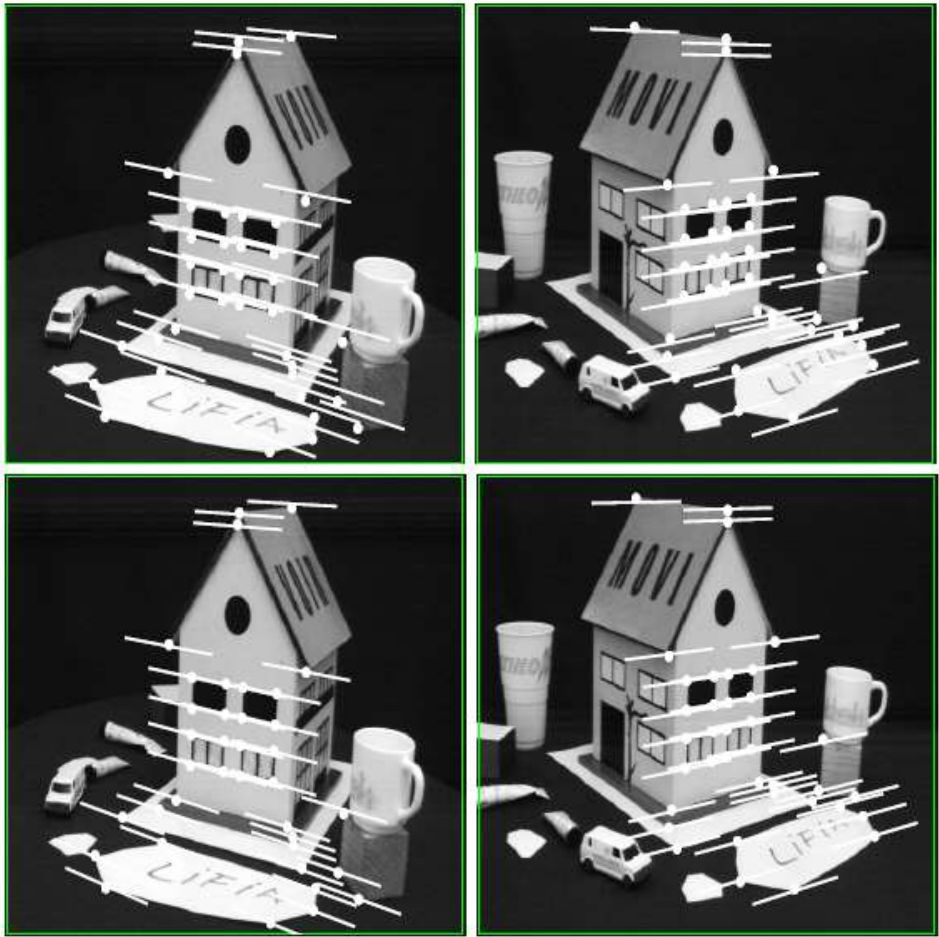
$$\sum_i (\mathbf{x}'_i{}^T \mathbf{F} \mathbf{x}_i)^2$$

- Nonlinear approach: minimize sum of squared *geometric* distances

$$\sum_i [\text{dist}(\mathbf{x}'_i, \mathbf{F} \mathbf{x}_i)^2 + \text{dist}(\mathbf{x}_i, \mathbf{F}^T \mathbf{x}'_i)^2]$$



Comparison of Estimation Algorithms



	8-point	Normalized 8-point	Nonlinear least squares
Av. Dist. 1	2.33 pixels	0.92 pixel	0.86 pixel
Av. Dist. 2	2.18 pixels	0.85 pixel	0.80 pixel

Seven-Point Algorithm

- Set up least squares system with seven pairs of matches and solve for null space (two vectors) using SVD
- Solve for polynomial equation to get coefficients of linear combination of null space vectors that satisfies $\det(F) = 0$

From Epipolar Geometry to Camera Calibration

- Estimating the fundamental matrix is known as “weak calibration”
- If we know the calibration matrices of the two cameras, we can estimate the essential matrix: $E = K'^T F K$
- The essential matrix gives us the relative rotation and translation between the cameras, or their extrinsic parameters
- Alternatively, if the calibration matrices are known (or in practice, if good initial guesses of the intrinsics are available), the five-point algorithm can be used to estimate relative camera pose

The Fundamental Matrix Song